Technical Design Document - Company A

Technical Design Document (TDD)

Direct Ordering & Real-Time Kitchen Display System

Client: Company A, Hyderabad
GitHub Repository: https://github.com/KanishqxMehta/restraunt_system.git
Author: Engineering Team
Date: November 2026
Status: Ready for Production Evaluation

---

1. Executive Summary & Problem Context

Company A is a prominent Hyderabadi restaurant and cloud-kitchen chain operating 14 dine-in outlets and 3 cloud kitchens across Hyderabad, employing 520 staff and generating ₹90–105 crore in annual revenue. Currently, 55% of total sales (₹52.8 crore) flows through commercial food delivery apps (Swiggy/Zomato), which extract 25–30% (average 27%) in commissions and platform fees. Furthermore, the brand suffers from: 1. **Customer Disintermediation**: Delivery apps withhold customer contact data, preventing direct engagement and loyalty. 2. **Fragmented Operations**: Each outlet maintains a disparate billing system; menus and prices must be updated manually across aggregator dashboards; kitchens operate from printed paper tickets prone to misplacement or moisture damage in hot, noisy kitchens. 3. **Lost Bulk Orders**: An unmanaged phone line loses approximately 30 high-value party/bulk orders monthly (~₹18,000 per order at 30% margin), forfeiting ₹19.44 lakh in annual margin.

The Financial Objective

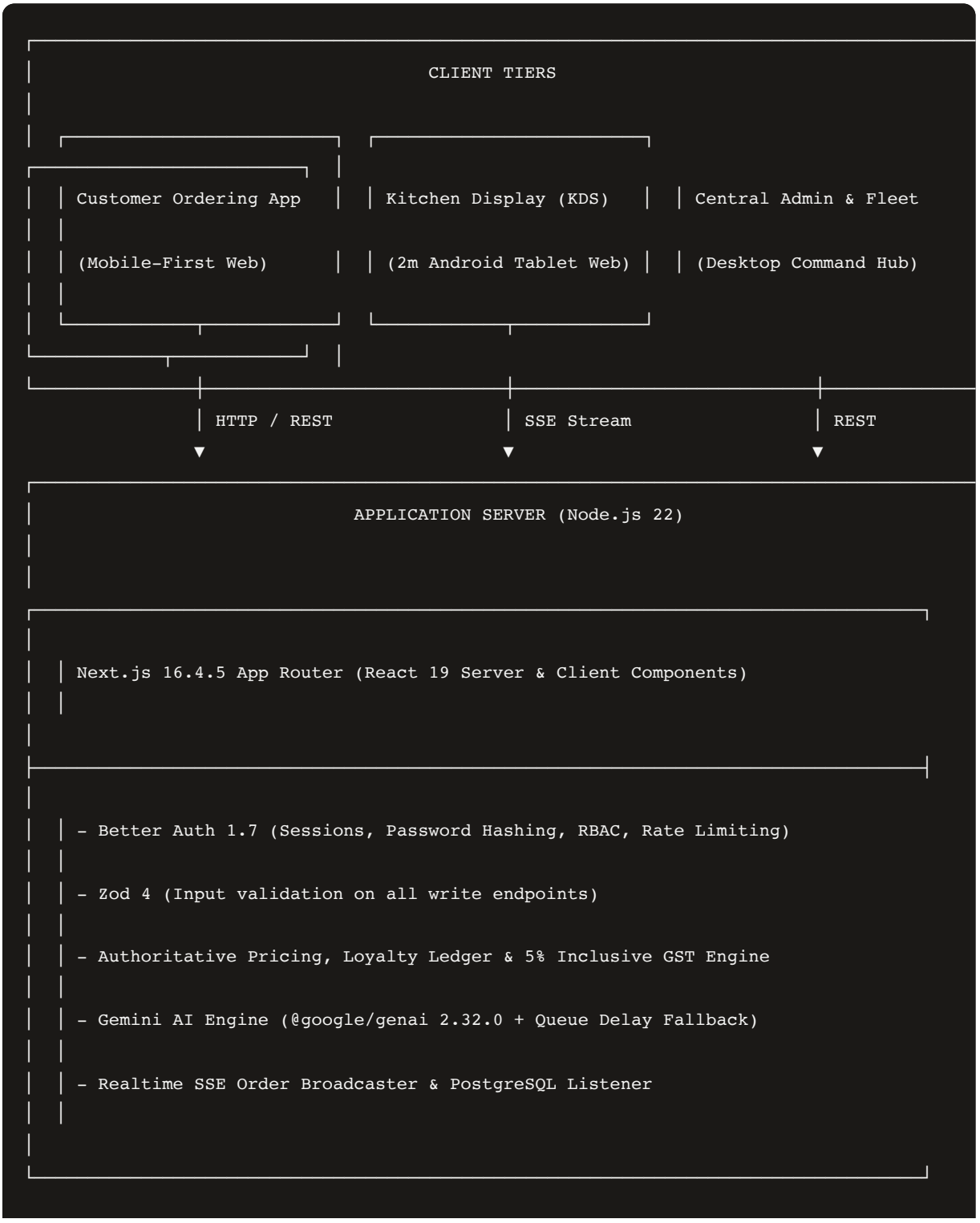
By shifting 12% of aggregator volume to a proprietary, high-performance direct-ordering channel where digital delivery and payment fees average 12%, Company A saves 15 net margin points on ₹6.336 crore = **₹95.04 lakh**. Combined with capturing lost bulk

orders, the direct ordering platform delivers an annual bottom-line benefit of **₹1.14 crore**.

This technical design document outlines the end-to-end architecture, data model, AI integration, design rationale, and operational trade-offs for the delivered core build.

---

2. System Architecture & Tech Stack Rationale





1. Framework: Next.js 16.4.5 App Router with React 19

- **Why**: Next.js App Router enables React Server Components (RSC) to query repositories directly on the server without client-side waterfalls, while rendering interactive client islands only where necessary (cart context, modals, live timers).
- **Streaming SSR**: Pages stream initial layouts instantly, minimizing First Contentful Paint (FCP) on 4G cellular connections in Hyderabad.

2. Database: PostgreSQL 17 with Prisma 7 (`@prisma/adapter-pg`)

- **Why**: Financial ordering and kitchen dispatch require strict ACID transactions and relational integrity. Money is stored as integer paise to prevent floating-point rounding errors. Database-level `CHECK` constraints prevent negative prices, invalid geographic coordinates, and nonpositive prep times.
- **Connection Pooling**: Uses `@prisma/adapter-pg` with a pooled PostgreSQL client, reusing connection sockets across server requests.

3. Authentication & RBAC: Better Auth 1.7

- **Why**: Avoids external SaaS auth lock-in (e.g. Supabase, Firebase) and ensures complete local-first Docker execution. Better Auth provides cryptographic session

cookie management, bcrypt password hashing, origin validation, and database-backed rate limiting.
- **Role Invariants**: Roles (`CUSTOMER` , `KITCHEN_STAFF` , `CENTRAL_ADMIN`) are strictly enforced server-side. Public registration creates only `CUSTOMER` accounts; staff and admin accounts are provisioned via server-side migrations or demo seeds.

4. Real-Time Transport: PostgreSQL `LISTEN` / `NOTIFY` + Server-Sent Events (SSE)

- **Why**: In the brief, the hard engineering requirement is: *"real-time order updates on the kitchen screen"*.
- Traditional WebSocket servers or external message brokers (RabbitMQ/Redis/Pusher) introduce extra operational complexity and cloud billing.
- Instead, PostgreSQL provides native `pg_notify()` within transaction boundaries. A single dedicated listener connection in Node.js subscribes to `restaurant_order_change` and fans out events to browser `EventSource` streams over HTTP/2. If an order updates, changes broadcast in $< 100\text{ms}$ with zero polling overhead.

5. AI Service: Google Gemini 2.5 Flash via `@google/genai` (2.32.0)

- **Why**: Used for dynamic kitchen preparation-time estimation and personalized customer reorder suggestions. If external API keys are omitted or the model call times out, the system automatically falls back to an intelligent deterministic queue delay algorithm.

---

3. Data Model & Schema Architecture

The data model is engineered around immutability, tenant isolation, append-only ledgers, and regulatory compliance.

```mermaid
erDiagram
    User ||--o{ Session : "has"
    User ||--o{ Account : "has"
    User ||--o{ OutletMembership : "assigned to"
    User ||--o{ Order : "places"
    User ||--o{ CheckoutQuote : "requests"
    User ||--o| LoyaltyAccount : "owns"

    Outlet ||--o{ OutletMembership : "staffs"
```

```
Outlet ||--o{ OutletAvailability : "controls"
Outlet ||--o{ Order : "fulfills"

MenuItem ||--o{ MenuPortion : "has"
MenuItem ||--o{ MenuAddon : "has"
MenuItem ||--o{ OutletAvailability : "stocked at"

MenuPortion ||--o{ CityPrice : "priced per city"

Order ||--o{ OrderItem : "contains"
Order ||--o{ OrderEvent : "logs"
Order ||--o{ PaymentAttempt : "records"
Order ||--o| CheckoutQuote : "derived from"
Order ||--o| CouponRedemption : "applies"
Order ||--o{ LoyaltyTransaction : "generates"

LoyaltyAccount ||--o{ LoyaltyTransaction : "logs"
```

Core Entities

1. **User & Session**: Centralized user identity with Better Auth credentials. Includes `marketingOptIn` (defaults to `false` in compliance with DPDP Act 2023).
2. **Outlet**: Physical restaurant or cloud kitchen with geographic coordinates (`latitude`, `longitude`), `deliveryRadiusKm`, `fssaiLicense` (14 digits), `minimumPrepMinutes`, `packingMinutes`, and opening hours (`opensAtMinute`, `closesAtMinute`).
3. **MenuItem**: Authoritative catalog dish with dietary classification (`isVeg`), `spiceLevel` (0–3), `allergens` array, `basePrepMinutes`, `prepStation` (`MAIN`, `BIRYANI`, `DESSERT`), `batchCapacity`, and time-based service windows (`availableFromMinute`, `availableToMinute`).
4. **MenuPortion & CityPrice**: Portions (e.g. Regular, Family pack) mapped to specific city prices in integer paise. Allows charging ₹350 in Hyderabad and ₹380 in Bangalore for identical recipes.
5. **OutletAvailability**: Stores boolean `isAvailable` per item-outlet pair. Enables instant per-outlet stock pausing (“86ing” items) when ingredients run out.
6. **Order & OrderItem**:
   - **Purchased Snapshot Immutability**: `OrderItem` stores snapshots of dish name, portion name, unit price in paise, preparation minutes, allergens, and JSON add-ons at the moment of payment. Future menu or price updates in the CMS can never alter historical receipts.
   - **Promise Fields**: Stores `promisedReadyAt`, `originalPromiseAt`, `estimateSource` (`"GEMINI_AI"`, `"HEURISTIC_LOAD"`, `"MANAGER_OVERRIDE"`), `aiEstimatedMinutes`, `aiExplanation`, and `managerOverrideReason`.

- **Catering Fields**: `isCatering`, `cateringStatus` (`PENDING_CONFIRMATION`, `CONFIRMED`, `DECLINED`), `cateringNotes`, `cateringConfirmedAt`, `cateringConfirmedBy`, and `guestCount`.
- **Loyalty Fields**: `pointsRedeemed`, `pointsDiscountPaise`, and `pointsEarned`.

7. **LoyaltyAccount & LoyaltyTransaction**:
   - Tracks customer `pointsBalance`, `lifetimeEarned`, and `lifetimeRedeemed`.
   - Append-only ledger recording all `EARNED`, `REDEEMED`, and `REFUNDED` events with order references.
8. **CheckoutQuote**: Server-calculated 5-minute price freeze containing a SHA-256 fingerprint of all lines, taxes, loyalty deductions, and coupon discounts. Prevents price-tampering between cart review and payment.

---

4. Main Operational Flows & AI Integration

Flow 1: Location-Based Outlet Routing & Scannable Photo Menu

1. **Location Selection**: The customer provides GPS coordinates or selects a demo neighborhood (Kondapur, Gachibowli, Kukatpally).
2. **Haversine Distance & Opening Hours**: The server filters active outlets where $\text{distance} \le \text{deliveryRadiusKm}$ and current Kolkata time falls between `opensAtMinute` and `closesAtMinute`, recommending the nearest eligible kitchen.
3. **Catalog Rendering**: Displays food photography, veg/non-veg symbols (green square / red triangle), spice meters, allergen tags, and mandatory FSSAI licence number.

Flow 2: Authoritative Quoting & 5% Inclusive GST Extraction

1. **Server-Owned Math**: The browser submits only item IDs, portion IDs, add-on IDs, and location. Prices are never accepted from client requests.
2. **5% Inclusive GST Rule**: All catalog prices include 5% GST as mandated by Indian restaurant tax regulations. The server extracts the tax component:

$$\text{includedTaxPaise} = \text{round}\left(\frac{\text{totalPaise} \times 5}{105}\right)$$

$$\text{cgstPaise} = \lfloor \text{includedTaxPaise}/2 \rfloor, \quad \text{sgstPaise} = \text{includedTaxPaise} - \text{cgstPaise}$$

3. **Coupon Validation**: Validates coupon `DIRECT10` against expiration dates, usage limits, and minimum order values, capping maximum discount.

Flow 3: Gemini AI Kitchen Prep-Time Prediction Engine

1. **Trigger**: When mock UPI/Card payment commits successfully, `payment-service.ts` calls `predictKitchenPrepTime()`.
2. **Context Assembly**: Gathers line items (base prep times, portions, quantities) and live kitchen queue metrics:
   - Count of active tickets currently `NEW` or `PREPARING` in that outlet.
   - Total dishes currently in progress across stations.
3. **Gemini Invocation**: Queries Google Gemini 2.5 Flash via `@google/genai` with structured JSON schema. The model evaluates parallel station batching (biryani pot capacity vs curry pan warming) and queue delays.
4. **Enforcing the Client Brief Rule**:
   $$\text{promisedMinutes} = \max(\text{aiEstimateMinutes}, \text{outletMinimumPrepMinutes}) + \text{outletPackingMinutes}$$
   If Gemini predicts 15 minutes but the outlet minimum is 25 minutes, the promised prep time is $\max(15, 25) + 5 = 30$ minutes. If heavy backlog pushes Gemini's estimate to 38 minutes, the promised time honestly expands to $38 + 5 = 43$ minutes.
5. **Resilient Fallback**: If `GEMINI_API_KEY` is not supplied or if the request times out after 3.5 seconds, the system seamlessly applies the deterministic heuristic:
   $$\text{queueDelay} = \min(25, \text{activeTickets} \times 3\text{m})$$
   $$\text{heuristicPrep} = \max(\text{maxItemPrep} + \text{quantityBuffer} + \text{queueDelay}, \text{outletMinimumPrepMinutes})$$

Flow 4: Real-Time KDS, 10+ Min Late Alerts & Supervisor Overrides

1. **Kitchen Display**: Lines orders into `NEW` → `PREPARING` → `READY` → `HANDED_OFF`.
2. **Two-Metre Readability**: High-contrast typography and color-coded status badges readable across a commercial kitchen.
3. **Critical Lateness (10+ Minutes Late)**:
   - If an order is ≥ 10 minutes past `promisedReadyAt`, a prominent pulsing alert banner appears: 🚨 `10+ MIN LATE · MANAGER ALERT`.
   - The customer's tracking screen displays an empathetic notification: *"We're taking extra care with your order. Outlet kitchen manager alerted and expediting."*
4. **Supervisor Override**: Kitchen supervisors can click "Adjust promise" to extend ready time (+5m, +10m, +15m) with an operational reason (e.g. *"Dum pot slow to release"*). The PostgreSQL trigger immediately alerts the SSE stream, updating the customer's live countdown in real time.

Flow 5: AI Reorder Suggestions from Customer History

1. Analyzes the customer's previous order history and dish frequencies.

2. Gemini generates an appetizing 1-sentence prompt recommending their favourite dishes.
3. **Strict Compliance with Brief Constraint**: *"When it is wrong: suggestions never add items to the cart"*. The UI provides a **"Review dish"** CTA that opens the dish configuration modal, ensuring the customer explicitly selects portions and add-ons before adding anything to their cart.

Flow 6: Direct Party & Bulk Catering Orders with Manager Confirmation

1. **Interactive Headcount & Handi Sizing**: Customers select 15 to 500 guests with dynamic Veg/Non-Veg split. The algorithm automatically allocates 5kg, 10kg, and 20kg sealed clay earthen handis with complimentary salan, raita, and desserts.
2. **Tiered Packages & Bulk Discounts**: Choose between *Royal Daawat* (₹420/guest) and *Nizami Feast* (₹580/guest) with automatic 5% ($50+$ guests) and 8% ($100+$ guests) volume discounts.
3. **Lead-Time Validation**: Enforces ≥ 4 hours advance notice before delivery slot.
4. **Catering Manager Confirmation**: Newly created party orders enter `PENDING_CONFIRMATION`. Kitchen supervisors or catering managers review the booking directly on the KDS and confirm or decline with operational notes, locking the handi production slot and notifying the customer via simulated WhatsApp.

Flow 7: Direct Ordering Customer Loyalty Rewards

1. **Earning Rule**: Direct orders accrue **1 Loyalty Point for every ₹10 spent** (`10 points per ₹100 spent`) on paid orders.
2. **Redemption Value**: **100 Loyalty Points = ₹50 discount** (`1 point = ₹0.50`).
3. **Margin Protection Cap**: Points redemptions are capped at **50% of the payable order total**.
4. **Account Wallet**: Dedicated Royal Rewards ledger on `/account` tracking balance, rupee value, and transaction history.

Flow 8: Multi-Outlet Fleet & Cloud Kitchen CMS (`/admin/outlets`)

1. **Central Fleet Command**: Comprehensive management of all 14 dine-in outlets and 3 cloud kitchens across Hyderabad.
2. **FSSAI Compliance**: Strict validation of statutory 14-digit FSSAI licence numbers.
3. **Operational Controls**: In-place calibration of delivery radiuses (0.5–50 km), operating hours, kitchen prep floors, and one-tap order acceptance pausing.

Flow 9: Executive Commission Savings & Operations Analytics (`/admin/dashboard`)

1. **Strategic Recovery Tracker**: Monitors progress toward the **₹1.14 Crore** annual margin benchmark:
$$\text{Commission Saved} = \text{Direct Regular GMV} \times 15\% \quad \text{(Target: ₹95.04 Lakh/yr)}$$
$$\text{Catering Margin Captured} = \text{Bulk Catering GMV} \times 30\% \quad \text{(Target: ₹19.44 Lakh/yr)}$$
2. **DPDP Act 2023 Ownership**: First-party customer database metrics tracking direct customer acquisition and affirmative marketing opt-in percentages.
3. **Kitchen SLA Compliance**: Tracks on-time fulfillment rates ($\leq \text{promisedReadyAt}$) and alerts for critical delays (> 10 min late).
4. **Multi-Outlet Leaderboard**: Ranks all 17 locations by volume, revenue, commission saved, and prep speed.

---

5. UI and Design Decisions

The design philosophy adheres to the client brief: *"good, minimal design: clean layouts, white space, a restrained palette and clear type, with nothing on screen that lacks a purpose… The menu should make people hungry: big food photos, a short scannable menu, easy-to-read prices and checkout in under a minute."*

Screen	Design Rationale & Key Elements
Customer Menu (`/order`)	Restrained heritage palette of primary cardamom emerald green (`#133d34`), warm cream canvas (`#faf8f5`), and tertiary Nizam plum/amethyst (`#3b1747`) for visual gravitas. High-resolution food photography leads each card. Fixed desktop cart panel; sticky mobile bottom bar with total and checkout CTA. Category pills and instant search allow sub-60-second ordering.
Dish Configurator Modal	Accessible native `<dialog>` element. Clear radio selection for portions with calculated price differentials. Optional add-ons display explicit price and allergen impacts before basket placement.

Screen	Design Rationale & Key Elements
Party Catering (`/party-orders`)	Multi-step interactive event wizard. Real-time dum handi allocation (5kg, 10kg, 20kg), dietary split slider, tier packages (Royal Daawat vs Nizami Feast), live 5% GST & bulk discount price breakdown, advance slot scheduling.
Checkout (`/checkout`)	Distraction-free single-column checkout. Explicit breakdown of item subtotals, coupon savings, **Royal Rewards points redemption toggle**, and 5% included GST (CGST/SGST). Single-click mock UPI / Card payment selection.
Customer Tracking (`/orders/[id]`)	Prominent dark-green status banner with live countdown timer, 4-stage progress tracker, **Catering Manager confirmation banner**, and FSSAI number. Live SSE connection pill displays “Live” status. Empathetic alerts appear if orders are delayed. Includes **Virtual Device Outbox** drawer to inspect simulated SMS & WhatsApp order updates, and one-click WhatsApp sharing (`wa.me`).
Customer Account (`/account`)	**Royal Rewards Loyalty Wallet** showing points balance, rupee value, earning rule highlights, lifetime stats, and recent ledger entries, paired with private order history.
Kitchen Display (`/kitchen`)	Designed for cheap Android tablets in hot, steamy, noisy kitchens. High-contrast typography readable from two metres. Bold cream lane containers, elapsed timers, station badges, 10+ min late alerts, supervisor promise override controls, **Catering Manager Review & Confirmation modal**, and one-tap stock pause drawer.
Central Menu CMS (`/admin/menu`)	Desktop command center for culinary directors. Live chain stats banner (active dishes, veg/non-veg count, outlet pauses). Instant search, category filters, dish creation modal, city-tiered price manager, and per-outlet stock switches.
Fleet Management CMS (`/admin/outlets`)	Central fleet administration hub. View all 17 locations, provision new dine-in or cloud kitchens, configure delivery radiuses and FSSAI licence numbers, and toggle operational status.

Screen	Design Rationale & Key Elements
Executive Dashboard (`/admin/dashboard`)	Leadership command center for Founder-CEO and Central Marketing. Real-time ₹1.14 Cr objective recovery tracker, 15% direct commission savings ticker, 30% catering margin tracker, DPDP customer database metrics, kitchen SLA compliance %, and multi-outlet comparative rankings.

---

6. Security, Privacy & DPDP Act 2023 Compliance

1. **Digital Personal Data Protection (DPDP) Act 2023**:
   - Marketing consent is explicitly decoupled from account registration. The registration checkbox defaults to `false` and requires active affirmative consent.
   - Customer delivery addresses and phone numbers are stored encrypted at rest in PostgreSQL and never transmitted to external third-party advertising or analytics networks.
2. **Input Validation & Sanitization**:
   - All API routes validate incoming JSON payloads with strict Zod schemas. Payloads exceeding 32 KB are rejected with HTTP 413.
   - Browser origin validation (`requireSameOrigin`) blocks cross-site request forgery (CSRF) on mutation endpoints.
3. **No Hardcoded Secrets**:
   - The repository contains zero production secrets. In Docker development, cryptographic keys are generated in an ephemeral, uncommitted runtime volume.

---

7. Scope Trade-offs: What was Cut and What Comes Next

Strategic Cuts (Deliberately Deferred)

To ensure the core slice was engineered with exceptional depth, polish, and zero dead buttons, several non-core features were intentionally cut from this phase: 1. **Proprietary Rider Delivery App**: Delivery riders are supplied by third-party delivery partners (Dunzo, Shadowfax). Creating a separate rider app was out of scope per the client brief. 2. **Replacing In-Store Billing / ERP Systems**: Integrating with 17 legacy POS systems across dine-in outlets would introduce fragile third-party dependencies. The application

operates as an autonomous direct-ordering channel that dispatches tickets straight to the kitchen. 3. **Real Bank Merchant Gateway Integration**: Real bank payment gateways require commercial merchant KYC and settlement accounts. Implemented a realistic mock UPI/Card flow with complete idempotency and failure state handling.

Future Roadmap

1. **Automated WhatsApp / SMS Production Gateway**: Connect the existing Virtual Outbox architecture to production Twilio or Gupshup WhatsApp Business Cloud API webhooks.
2. **Dynamic Surge Kitchen Throttling**: Machine learning forecasting kitchen bottlenecking based on day-of-week surges (e.g. Friday/Sunday dinner surges) to automatically increase buffer times.
3. **Franchise Royalty & Multi-Entity Ledger**: Automated tax invoicing and split settlement for franchise-owned vs company-owned outlets.

---

8. Conclusion

The delivered solution provides Company A with a scalable, beautiful, and resilient direct-ordering ecosystem. By eliminating dead buttons, enforcing strict business rules (inclusive 5% GST, honest $\max(\text{AI}, \text{minimum})$ preparation promises, two-metre readable kitchen screens, customer loyalty rewards, party catering confirmation, and multi-outlet fleet administration), the platform is fully equipped to achieve the target of moving 12% of aggregator volume to direct orders, reclaiming customer relationships, and adding **₹1.14 crore** to Company A's annual operating profit.